

Differentially Private Obfuscation of Power Grid Dynamics

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Motivation

Power system dynamics governs how the grid responds to disturbances and rapidly changing conditions. Accurate dynamic models are imperative for resilient control and post-event analysis of blackouts. Yet, releasing real-world dynamic models is hindered by privacy and cybersecurity concerns. For example, it took two years for the dynamics of 2025 Iberian Peninsula blackout to be released.

Research Questions

- How to enable the secure release of power system dynamics models?
- How to ensure the fidelity of the released dynamics models?

Grid Dynamics and Sensitive Parameters

The linearized transmission system dynamics model is posed as

$$\mathbf{M}\dot{\boldsymbol{\omega}} = \mathbf{K}\boldsymbol{\delta} + \mathbf{p} - \boldsymbol{\ell} - \mathbf{D}\boldsymbol{\omega} \quad (1a)$$

$$\dot{\boldsymbol{\delta}} = \boldsymbol{\omega} \quad (1b)$$

$$\dot{\mathbf{p}} = -\mathbf{T}(\mathbf{p} + \mathbf{R}\boldsymbol{\omega}) \quad (1c)$$

State variables: frequency $\boldsymbol{\omega}$, voltage angles $\boldsymbol{\delta}$, mechanical power output $\mathbf{p}$.

System design parameters: generator inertia $\mathbf{M}$ (sensitive), nodal damping $\mathbf{D}$ (sensitive), weighted Laplacian $\mathbf{K}$ (sensitive), inverse of governor/turbine time constant $\mathbf{T}$, droop constants $\mathbf{R}$.

Disturbances: load deviation $\boldsymbol{\ell}$

Topology Obfuscation via Kron Reduction

Separate the system nodes into two sets, boundary nodes γ that includes all generator nodes, and interior nodes β whose damping are ignored, the dynamics model becomes

$$\begin{bmatrix} \mathbf{M}_\gamma & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\omega}_\gamma \\ \boldsymbol{\omega}_\beta \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{\gamma\gamma} & \mathbf{K}_{\gamma\beta} \\ \mathbf{K}_{\beta\gamma} & \mathbf{K}_{\beta\beta} \end{bmatrix} \begin{bmatrix} \boldsymbol{\delta}_\gamma \\ \boldsymbol{\delta}_\beta \end{bmatrix} + \begin{bmatrix} \mathbf{p}_\gamma \\ \mathbf{0} \end{bmatrix} - \begin{bmatrix} \boldsymbol{\ell}_\gamma \\ \boldsymbol{\ell}_\beta \end{bmatrix} - \begin{bmatrix} \mathbf{D}_\gamma & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\omega}_\gamma \\ \boldsymbol{\omega}_\beta \end{bmatrix}$$

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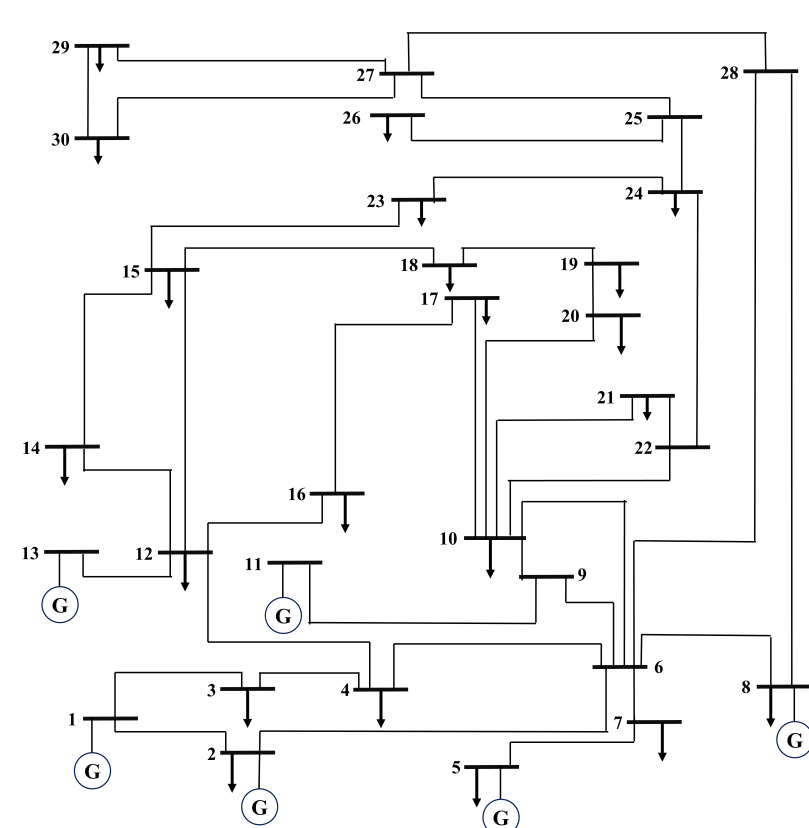
Kron reduction renders a reduced dynamic model via Gaussian elimination

$$\mathbf{M}_\gamma \dot{\boldsymbol{\omega}}_\gamma = \mathbf{K}_{\text{red}} \boldsymbol{\delta}_\gamma - \mathbf{D}_\gamma \boldsymbol{\omega}_\gamma + (\mathbf{p}_\gamma - \boldsymbol{\ell}_\gamma) - \mathbf{K}_{\text{ac}} \boldsymbol{\ell}_\beta \quad (3a)$$

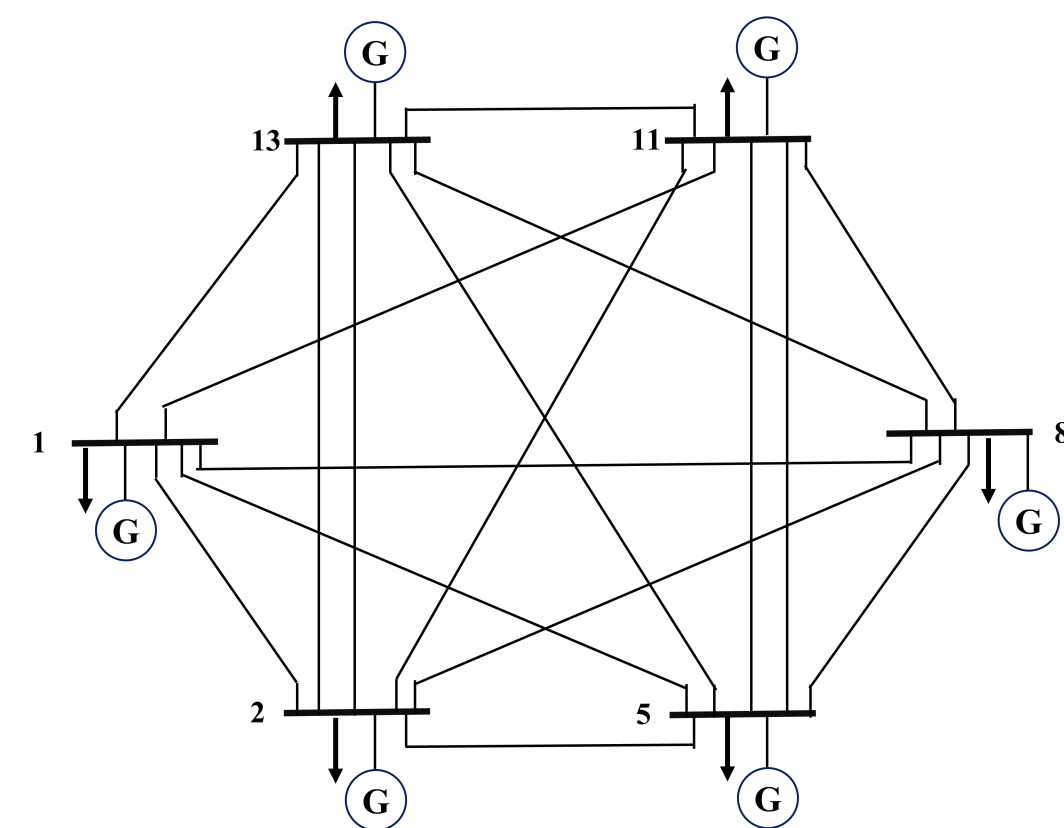
$$\dot{\boldsymbol{\delta}}_\gamma = \boldsymbol{\omega}_\gamma \quad (3b)$$

$$\dot{\mathbf{p}}_\gamma = -\mathbf{T}_\gamma(\mathbf{p}_\gamma + \mathbf{R}\boldsymbol{\omega}_\gamma) \quad (3c)$$

where $\mathbf{K}_{\text{red}} = \mathbf{K}_{\gamma\gamma} - \mathbf{K}_{\gamma\beta}\mathbf{K}_{\beta\beta}^{-1}\mathbf{K}_{\beta\gamma}$ is the reduced coefficient matrix, and $\mathbf{K}_{\text{ac}} = -\mathbf{K}_{\gamma\beta}\mathbf{K}_{\beta\beta}^{-1}$ is the accompanying matrix that maps the load disturbance of the reduced nodes onto the boundary nodes. The parameters of this model are stacked into a vector $\boldsymbol{\theta} = (\mathbf{k}, \mathbf{m}, \mathbf{d})$, where $\mathbf{m}$ and $\mathbf{d}$ are diagonals of $\mathbf{M}_\gamma$, $\mathbf{D}_\gamma$, and $\mathbf{k}$ reconstructs $\mathbf{K}_{\text{red}}$, $\mathbf{K}_{\text{ac}}$.



(a) Original IEEE 30-bus system



(b) Kron-reduced system

Parameter Obfuscation via Differential Privacy

The goal of DP algorithm is that, given a vector $\boldsymbol{\theta}$ of model parameter, it produces synthetic counterpart $\tilde{\boldsymbol{\theta}}$ that yields similar frequency dynamics yet does not disclose the original parameters up to prescribed privacy guarantees.

Definition 1 (Adjacency). *Two vectors $\boldsymbol{\theta}, \boldsymbol{\theta}' \in \mathbb{R}^n$ are α -adjacent if they are different in one item by at most α .* ◀

Definition 2 (ϵ -DP). *The Laplace mechanism is called ϵ -DP if, for any outcome $\tilde{\boldsymbol{\theta}}$ in the range of the mechanism, and any two α -adjacent vectors $\boldsymbol{\theta}$ and $\boldsymbol{\theta}'$, the ratio of probabilities of observing the same outcome is bounded as*

$$\left| \log \left(\frac{\Pr[\boldsymbol{\theta} + \text{Lap}(\alpha/\epsilon)^n = \tilde{\boldsymbol{\theta}}]}{\Pr[\boldsymbol{\theta}' + \text{Lap}(\alpha/\epsilon)^n = \tilde{\boldsymbol{\theta}}]} \right) \right| \leq \epsilon,$$

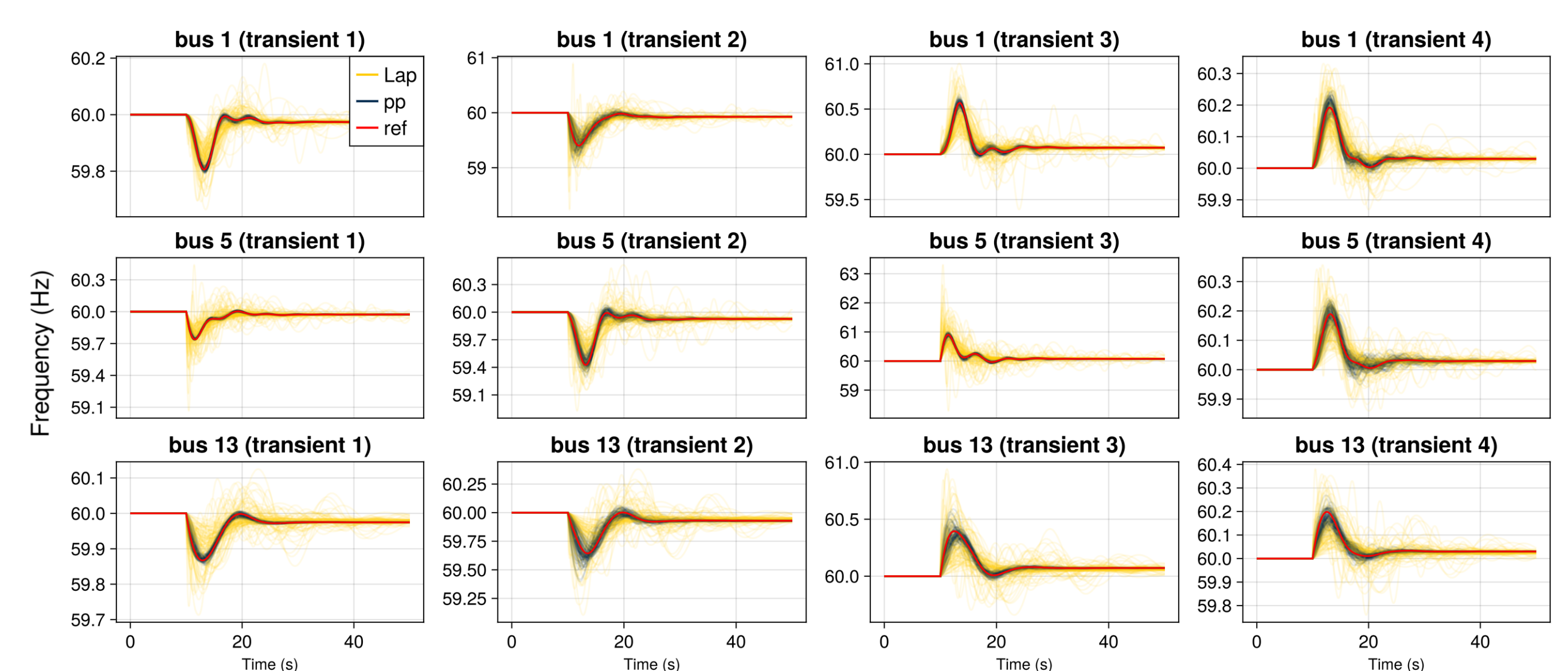
To ensure DP guarantee, we first do $\tilde{\boldsymbol{\theta}}^0 = \boldsymbol{\theta} + \text{Lap}(\alpha/\epsilon)^n$. Since the noise will decrease the fidelity of the model, we have the post-processing optimization with ODE constraints

$$\underset{\boldsymbol{\theta}, \mathbf{x}}{\text{minimize}} J(\mathbf{x}, \boldsymbol{\theta}) := \int_0^T \|\mathbf{H}\mathbf{x}(t) - \tilde{\boldsymbol{\omega}}(t)\|_2^2 dt + \rho \|\boldsymbol{\theta} - \tilde{\boldsymbol{\theta}}^0\|_2^2 \quad (4a)$$

$$\text{subject to } \dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \boldsymbol{\theta}) \quad : \boldsymbol{\lambda}, \quad \mathbf{x}(0) = \mathbf{0}, \quad (4b)$$

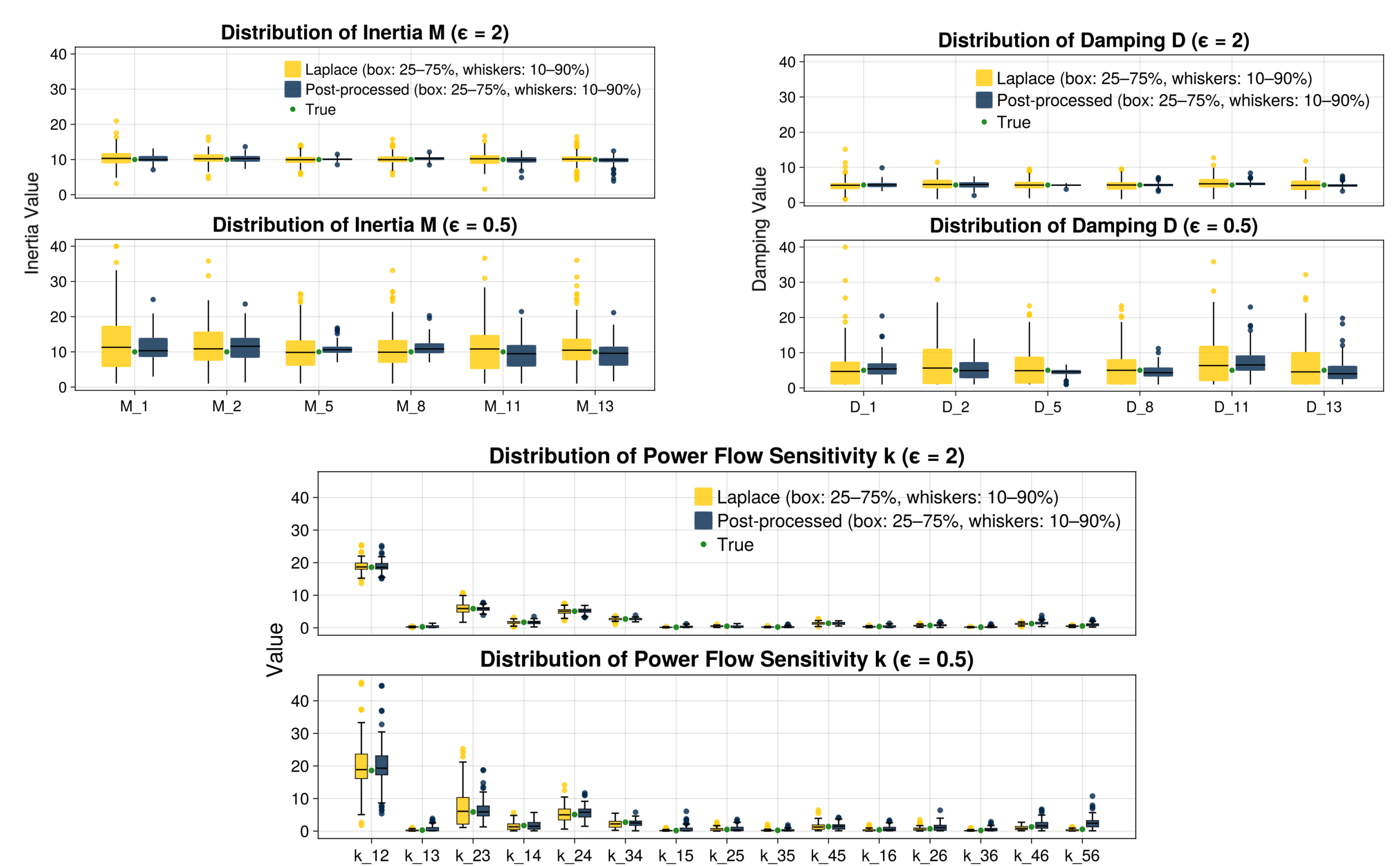
where $\boldsymbol{\theta}$ is the model parameters to be released, $\mathbf{x} = (\boldsymbol{\omega}, \boldsymbol{\delta}, \mathbf{p})$ the system states, and $\tilde{\boldsymbol{\omega}}$ the publicly known trajectory. The optimization is solved via gradient descent, where the gradient is computed using the adjoint method.

Results



Distribution of frequency trajectories with Laplacian noise only and after post-processing ($\epsilon = 1$)

- Lap perturbation alone (yellow) yields frequency trajectories that deviate substantially from the true one (red). Post-processing restores fidelity (blue).



- As ϵ decreases (i.e., stronger privacy protection), more noise is injected, leading to a wider spread in the distributions.
- Some parameters that changes a lot after post-processing (e.g., M_1, D_5, k_{12}, k_{23}) means they are essential for restoring true frequency trajectories.

Conclusion

- This work extends DP from steady-state grid optimization data to grid dynamics, protecting the dynamical model itself while preserving its frequency response.
- Stricter privacy requirements increase frequency trajectory deviation, yet post-processing consistently recovers fidelity across all privacy levels.

